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The Long Way Round

Autobiographical essays by Sungmin Kim

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Episode 1

The Boy Who Read Number Plates

> *Everyone named in this essay gave their permission in advance.*

Number Plates

My parents tell me that, even as a small child, I was fascinated by the numbers on car number plates. I think my love of math has been with me ever since. I was born in Daejeon on 9 February 1991. We lived in Dunsan-dong, then moved to Wolpyeong-dong, and I went to Seongnyong Elementary School.

In the fourth grade, in Daejeon, I was chosen as a representative for a mathematics competition and won a prize. In October of the fifth grade, in the second semester, we moved to Seoul and I transferred to Eunjeong Elementary School. There too I was put in the gifted class for math, and I placed highly in a nationwide competition. I went on to Mogil Middle School and graduated from Daeil High School, a general high school in Seoul. In math, at least, I was always a strong student.

Written down like that, it reads like the story of a child born for it. It is true that I liked math and was good at it. But the two habits I now teach my students, writing by hand and building the concept first, did not come from being good. They came from being wrong.

Learned by Being Wrong

Even a strong student can make mistakes when working everything out in their head. Making those mistakes myself was what taught me to write my work down.

I once did badly in a statistics exam in middle school. I found the subject dull, neglected it, and sat the exam without properly understanding the concepts. The problem was my preparation, not my ability. That experience taught me to take concepts seriously. From then on, I kept practising until I could use a concept in a problem, not just recognise it.

I did not realise all at once, on some particular day, that math is a matter of method and not of the head. Exploring mathematics freely and preparing for an entrance exam are different tasks. You have to solve a great many problems within a fixed time. It makes sense to develop a method and organise your study around it. That idea began when I was helping friends as a student, and years of teaching made it clearer.

My Teacher, and My Parents

At the academy I attended in high school there was a teacher, Mr Choi Kyung-jae. He had studied mathematics education at Seoul National University and had taught math for a long time, and above all his own ability in mathematics was outstanding. He drew out a great deal of what was in me. Later he asked me to tutor his son. The teacher I had learned from was trusting me to teach his own child. He now runs an online community on Naver called Areunyuga.

I never had private tutoring or took online lectures. That academy was the strongest influence on my school years. Learning from such a capable teacher helped me develop further.

My parents supported me completely. Perhaps that was because I was doing well. But had they pushed me to do better, I would have rebelled. I was that kind of child. So this is what I would say to parents: understand your child, believe in them, and encourage them. Looking back, though, I do wish my parents had set clearer rules for everyday life. Perhaps they held back because I was so rebellious.

As a teenager, nothing frightened me. It was an age that did not know fear. A little of that is still in me.

The Long Way Round

The road from high school to Cambridge was not a straight one. I entered university in 2009. That May I started a part-time job in an academy's study hall, taking students' questions. It was my first work as an instructor. Then came military service. Before my discharge, I saw a notice for the first intake of YMS, the UK's Youth Mobility Scheme working-holiday visa. I wanted to see a wider world. I applied, and I was accepted. In Britain I did all sorts of work, and among them was the work I had kept doing all along, teaching math at an academy. There I met a student preparing for the mathematics course at Cambridge. Watching that student, I thought: I could do this too.

So I sat the exams. As for a subject, I thought mathematics suited me better than engineering. What counted for a great deal, I think, was scoring S, Outstanding, across the board in STEP, the examination taken by applicants to mathematics at Cambridge. And so I entered Trinity College to read mathematics, on a scholarship that covered full tuition and living costs as well.

One Sentence

If the person I am now could send just one sentence to the teenager I was, I would write this: "You are going to have a great many experiences later on. Don't let studying be all you do."

If you ask whether I have ever felt that I disliked math, the scene that comes to mind is from much later. It was at Cambridge, when I had to study a subject that did not suit me. University mathematics was different again. That is the story of the next episode.

Episode 2

Among People Born for It

> *The classmates who appear in this episode are not named.*

A Subject That Did Not Suit Me

It was at Cambridge that I felt I disliked math. It was when I had to study a subject that did not suit me. University mathematics was different again.

I first felt that difference in a supervision during my first term. A supervision is a class in which a few students sit across from a supervisor and talk through the problems they have worked on. Until then I had been someone who concentrated on solving problems, and nothing else. Korean math is built around multiple-choice and short-answer questions. What I struggled with was sustaining a written argument through an open-ended problem.

Mathematics at Cambridge is taken in stages: Part IA, Part IB, and Part II. Complete them and you receive a BA, and some years later, by convention, the MA. That is how I finished the Mathematical Tripos. This episode is the story of those years.

The People Who Were Mad About It

I remember the first time I saw a classmate who was said to be "born for mathematics." Looking back, it was not that this classmate was born for it. It was a passion for mathematics beyond the ordinary. In the exam period, even over meals, the talk was still mathematics. Simply mad about it.

That friend was studying mathematics at every hour of the day. Cambridge had members of national IMO teams, the International Mathematical Olympiad, from country after country. And yet the person who came top was not a gold medallist. Natural ability alone could not explain what I saw. The way they approached a problem, and their attitude to the work, mattered enormously.

All-Nighters

I was a crammer by nature. Even so, in that short stretch I worked through a great many past papers. Before exams, being young, I stayed up almost every night. I was once caught sleeping in the library of the mathematics building, the CMS. I was called in by someone senior, whether the dean or the warden I could not tell you, who asked me why I had done it, and whether I had no home to go to. I said it was because I had been studying. The example sheets I went over again and again.

I am not recommending cramming. But even in that short time, what I did came down to two things: solving past papers, and solving the example sheets again. That is how I got a First in Part IA and became a College Scholar of Trinity College, a scholarship awarded for academic distinction.

The Hardest Part

The hardest part, as you would expect, was the exams. At Cambridge a whole year's work is examined all at once, in one exam period. Just before graduation, for personal reasons, there was a year's gap in my studies, and catching up again when I came back was hard. Long hours bent over books also brought on the early symptoms of a disc problem in my neck. I had a procedure for those symptoms recently. Loneliness never troubled me; it is not in my nature, so on that front I was fine.

Newton's Book

If I had to name one place I remember, it would have to be the library that holds Newton's book. Newton was a Trinity man. His book is still there in that library today.

What I Brought Back, and What I Left Behind

There is something I learned at Cambridge that I now use, unchanged, in a grade-10 classroom. Memorise a definition accurately, understand what it means, and apply it to a problem. Connecting those three steps deepens mathematical understanding.

There are also things that cannot be brought back as they are. A university class at Cambridge and an entrance-exam classroom in Korea are different environments. Population density and student-to-teacher ratios differ too. The Cambridge approach cannot simply be transplanted unchanged.

The Teaching Side

My path had leaned one way from early on. I had been teaching math for a long time, and I had felt that this work suited me. At first I meant to teach at a university. So after my first year I took the GRE Mathematics Subject Test with postgraduate study in the United States in mind. It was the mathematics test, not the general GRE. I scored in the 99th percentile, the top one percent. But for personal reasons I did not apply to graduate school. As soon as I graduated, I came back to Korea and began teaching math.

One Sentence

If I sent one sentence to my Cambridge self, it would be this: “All that time you spent going out to tutor, you should have spent a little more of it with people.”

Cambridge showed me how much method and attitude matter. Results were not determined by natural ability alone. I wanted to find out with my own hands whether that lesson would hold just as well in the unfamiliar, unforgiving world of Korean entrance-exam mathematics. So I, by then teaching students, became a candidate again. That is the story of the next episode.

Episode 3

The Day I Sat the Exam Again

> *The other people in this episode have been written so that they cannot be identified.*

The Year I Came Back

After graduating I came back to Korea. There was no particular reason. I had graduated, so I came home. As soon as I was back I was teaching entrance-exam math of every kind, in one place after another. I had brought a Cambridge degree with me, but the work I did was the same work I had been doing since before I entered university: teaching math at an academy. Only this time it was my profession.

Why Grades 10 and 11

People who hear my background ask. With a record like that, shouldn't you be teaching the top students in their final year, or a medical-school essay class? Why grade-10 and grade-11 school exams? I wrote the first answer in the prologue of my book: I wanted to prove something. The second answer is this. I wanted to feel for myself, in the classroom, step by step from grade 10, what a grade-12 student is up against. The difficulties of grade 12 do not begin in grade 12. In the prologue I wrote about a dam. A small crack that opens silently in grades 10 and 11 brings down the whole dam by the time grade 12 arrives. To see the place where that crack opens, you have to be in a grade-10 classroom.

What I Told No One

There was one reason I decided to become a candidate again. I wanted to know whether I truly had a method for mathematics. If someone who tells students about a method has never once tested that method with his own hands, how light those words would be. I told no one. I simply went and did it, alone. I prepared by working through past papers, and nothing else. There was nothing much to be afraid of. Had I failed, it would quietly have become something that never happened. I had told no one, after all.

The Exam Hall

Yonsei University's essay examination for 2022 entry was held on Saturday, 2 October 2021, ahead of the CSAT. There was nothing special about the morning of the exam. I went on time, carrying my pens and my admission slip. Among the grade-12 candidates sat a thirty-year-old instructor.

There were four mathematics questions. The first had you roll a die, use the number shown to build an isosceles right triangle, and find the expected number of lattice points on its boundary and inside it. The last had you draw line segments inside a triangle according to given conditions to form polygons, and then show, using sums of vectors, that the number of polygons meeting each of the three sides is the same. Problems you had to write out to the end, over a long stretch. The very kind that had stopped me in my first supervision at Cambridge.

While I was writing my answers, one thought came to me: ah, I'm going to get in. The answer sheet of the candidate next to me was almost blank. As I wrote in the prologue, the odds were more than a hundred to one. When the exam was over, I probably went straight back to work.

The Chinese Restaurant

I found out that I had been admitted on my own, in a Chinese restaurant, over jajangmyeon and sweet-and-sour pork. A first-round admission to Dentistry, through the essay-based track. I finished my meal. I told my family. Good, I thought, and that was all. I did not tell my students. Some of them saw what I had written on my website and said it was impressive. That was all there was to it. I had not done it to impress anyone.

One Hundred, Three Times

After the essay exam came the CSAT. On the CSAT for 2022 entry I chose Calculus as my elective and scored a raw 100. In Probability and Statistics I scored 100 on the September KICE mock exam for 2023 entry, and in Geometry on the September KICE mock exam for 2024 entry. All three subjects, a raw score of 100. Those three score certificates record what happened when I tested my method in the exam hall.

Sitting the exam among students still in school, I saw something. The age gap, of course, I felt. What caught my eye more was that the students could not move on from a question they did not know. When the time is gone you have to move on, but they stayed where they were stuck. It is the shortest road to ruining an exam. I saw it not through an instructor's eyes but from a candidate's seat. That experience also helped me explain exam strategy back in the classroom.

University, and Pharmacy

I did enrol in Dentistry. But teaching was my work, and my studies kept being put on hold. The academic load was too heavy to do both. So for 2024 entry I sat the essay exam again, meaning to move to Pharmacy, where the load is comparatively lighter. Once again it was a first-round admission. I did not enrol in Pharmacy. You cannot be registered at two places at once. I had tested the same method again, on different problems in different years.

After the Proof

If you ask what changed once the proof was done, I answer with the sentences I wrote in the prologue. Not because I am some born genius. I had simply worked out, precisely, the pattern of decoding that Korean math exams demand. Whether it is pure mathematics at Cambridge or the school exam you will sit tomorrow, mathematics is not a talent. It is a craft of decoding, and a craft is won by training.

Being able to solve a problem myself and helping a student solve it independently were different tasks. Showing the answer was not enough; I had to help students work through it with their own hands. That is the story of the next episode.

Episode 4

Where Students Quietly Give Way

> *The student stories in this episode are composites of many students. None of them is any one person.*

The Dawn Shuttle

In 2024 I took a class in the medical-track division of Gangnam Daesung Boarding Academy, a class built around the academy's own Gangdae materials. I would get up before dawn, board the shuttle at Gangnam Daesung, and ride out to the boarding medical-track campus in Icheon. The name says it all: these are students who move into a dormitory and stake a year on getting into medical school. The days I stood in front of them began before daybreak.

In Front of the Camera

In Daesung MyMac's live classes, Daeinra and the Gangdae K Final, I taught the practice-exam classes that tie everything together in the second half of the year. A classroom and a camera are different. In a classroom you can see the student. You can see the moment a hand stops, the moment the eyes leave the problem, and that is when you go over. In front of a camera you cannot see any of that. There is no way to check, right there, whether the students are keeping up. That was the hard part. Over my time teaching online I thought about this problem for a long while, and I built several solutions for it. Those I will make public separately, when the time comes.

Where They Give Way

I have seen many students give way. What follows is a story made by combining many of them. The place where they give way is usually one of three.

The first is attitude. "I'll just retake it next year anyway." There is still time to prepare, but this student's mind has already moved on to next year. Pessimism is not laziness. The hand keeps moving, but the head has given up first. This student is not solving problems. This student is passing time.

The second is environment. There are friends around who do not work hard, and they go off together to do nothing but play games. I have also seen students' grades improve while spending time with friends who study hard. Who you spend time with affects your studying too.

The third is method. It is the most common, and the quietest. The student who works through three problem books, keeps an error notebook, sits through every online lecture that is supposed to be the best, and whose score does not move. Not lazy. Not slow either. Simply pushing, diligently, against a wall, with a wrong method that nobody ever properly corrected. The prologue of my book was written for this student.

The Student Who Comes with a Question

Students' questions taught me something: even high scorers have gaps in their understanding. Skip an explanation because a student is doing well, and you may miss that gap. Scores alone cannot tell me where to pitch a lesson.

So when I open a class, I briefly ask about the prerequisites for that lesson. If they say they know it, I ask again whether they really do. For a lesson on exponential functions, say, it goes like this. "Do you know the laws of exponents?" "Yes." "What is two to the power of zero?" "One." "Why is it one?" If a student cannot explain why, we need to distinguish between memorising the rule and understanding it. Giving a familiar answer is not the same as applying the idea to a new problem.

A Lesson That Failed

There have been lessons that failed too. It was when I first took on the medical-track division. I went at my own pace. Even the top class in the division said it was too fast. If even the top class found it too fast, my own pace was the wrong measure. A lesson has to leave students enough time to work things out with their own hands. I learned that then.

The Three Questions Parents Ask

In consultations with parents, three questions come up again and again. "Our child has a good head, but…" I tell them that math is not done with the head. That is the point of the book's title. "If only our child would just study…" Worrying that a child is not studying changes nothing by itself. They need a routine that helps them start and keep going. That, I tell them, I will show them. "They get the ones they know wrong." Reliability is part of ability too. Knowing something includes being able to use it without getting it wrong. All three answers are in *Don't Solve Math in Your Head*.

When They Refuse to Write by Hand

Some students will not write by hand no matter how often you tell them. I do not force it. They have to get it wrong themselves before they know. That is how I learned it too. I only hope that the exam they get it wrong in is not one that matters.

The moment a student changes does not show up first in the score. It shows up first in behaviour. The student who only ever followed with the eyes starts writing by hand; the student who only ever looked at wrong answers starts tracing the examiner's intent even in the problems they got right. Those are the changes I look for before an improvement in scores. Study can produce different results only when the way a student works begins to change.

Repeaters, and Grades 10 and 11

At Daesung Academy's full-time class for repeat-year students, I taught CSAT repeaters as well. Many repeaters assumed they knew material because they had studied it before, yet gaps appeared when they tried to solve a problem. In contrast, I saw grade-10 and grade-11 students acknowledge what they did not know and learn eagerly. That was when I saw them make real progress. One more reason was added to my reasons for going to grades 10 and 11.

Reduced to a single line, what the classroom taught me is this: a student grows when they know that they do not know. Now that method had to be carried outside the classroom. That is the story of the next episode.

Episode 5

Before I Take the Stage

> *The other contestants and the judges in this episode are not named.*

The Banner

It was the second half of 2021, around the time I was preparing for the essay exam and the CSAT. I saw an advertising banner on the web. MathKorea, the first open selection contest for mathematics instructors, held jointly by Daesung MyMac and Daesung Academy. I was an instructor, so I applied. There was no grand reason. There was a banner.

Four Gates

The selection had four stages: documents, a written test, a recorded lecture, and then, on site, a lecture solving a problem in front of the panel together with an in-depth interview. So this is how thoroughly they choose, I thought. This was different from sitting the essay exam as a student. I was being assessed on how I taught the problem as well as on solving it.

The Killer Problem

The last gate was the live demonstration lecture: solving one killer problem in front of the panel. It was a calculus problem. As I recall, we were given the problem about a day in advance. Before me sat more than ten people, the company's chief executive among them. I planned the explanation so that more students could follow it, not just those already strong at math. Judges instead of students, a judging room instead of a classroom. Solving the problem was the same.

The Result

The awards ceremony was held on 29 December 2021. First place was not mine. I was one of the two runners-up. Five million won in prize money, and a trophy. The result came by phone and by text message. Riding on first place were one hundred million won and the launch of an online course on Daesung MyMac. That stage, at that time, was not mine.

When I heard the result, two sentences went through my mind. I have been chosen by the country's top education company. Work hard. Students may have their own view of which company is best, but that was how I felt that day. Not being first was the question that came after.

The Stage

My Daesung MyMac course was confirmed in early 2026, in discussions with the company's executives. From December 2026 I am scheduled to teach grade-10 and grade-11 school-exam mathematics and CSAT mathematics: Common Mathematics 1 and 2, Algebra, Calculus I, and Probability and Statistics. Beginning with the 2028 CSAT, mathematics is examined only on Algebra, Calculus I, and Probability and Statistics. What students learn in grades 10 and 11 provides the foundation for the CSAT. That is why it makes little sense to treat school-exam study and CSAT preparation as separate tasks. Personally, I am also thinking of extending to Calculus II and Geometry. Several years on from that awards ceremony, I am now preparing for that stage.

These days I move between three places: my office, the head office where the filming is done, and home. The work carries on after I get home. It will be like this until December.

A Worry, and One Thing I Will Not Do

As the launch approaches, my biggest worry is whether students will enjoy my classes. In front of the camera, after all, I cannot see their faces.

There is also one thing I have decided not to do: exaggerate. I will not claim an effect that is not there, and I will not say that method can take the place of effort. Teaching a useful method is not the same as promising better grades without effort. I will keep that distinction clear.

The Team

Trinity MathsLab is not an old company. I set it up in the first half of 2026, because I needed a team to launch the online course with. I had built a way of seeing the student who cannot be seen from behind a camera, and I needed people to carry it out with me. The name comes from the college where I learned mathematics. The team has been trimmed to its core people and the way we work streamlined. I will not disclose the team's size. Between us we divide the work of making the lessons, filming them, and looking after the students.

One Sentence

If I sent one sentence to myself before I take the stage, it would be this: “You can do this. You are doing it already. Don’t give up. Keep going just as you are.”

Before taking the stage, I wrote a book first. That is the story of the next episode.

Episode 6

The Method Is in Your Hands

Why I Wrote the Book

The reason I decided to write a book is simple. I wanted more students to get more out of the lessons I had worked so hard to film. I hoped that learning good study habits from the book would help students get more out of a lesson. The habit of writing by hand, the habit of building the concept first, the habit of going back over even the problems you got right. What cannot be said in every lesson, I put into one book.

Don't Solve Math in Your Head. Thirteen sections, readable in Korean and in English. The full text is free at mathskim.com, on a phone, with nothing to sign up for. That is also why I made it free: I wanted more students to read it.

Is It a Well-Written Book?

One question stayed with me while I wrote, and it is still with me now: is this book really any good? The answer will come not from me but from a student who has followed this book for one semester.

Hyunwoo, the student in the book, is not a real student. Because of privacy, I could not write about an actual one. So I gathered what I had seen in many students into one. The student stories in this memoir were made the same way.

Two Sentences

My one sentence for parents is this: "Believe in your child, but set clear rules for everyday life." The first half is what my parents did for me; the second half is what I felt was slightly missing. I wrote as much in the first episode.

My one sentence for students is this: "Do your best, so that you leave no regret for the person you will be later." I was a child who did not know fear, and a crammer by nature, and even so, in that short time, I solved past papers by hand. Even if things do not go well, I hope you can say you tried everything you could.

Ten Years On

What will I be doing ten years from now? Let me make one joke. In this line of work there is a word, ilta, for the number-one instructor. I have even heard of someone, one step above that, speaking of a “zero-ta.” I would like to be the “minus-one-ta.” That is a joke. Half of it, anyway. But I mean it when I say that, ten years from now, I still want to be in a grade-10 classroom, telling students to write by hand.

An Honest Wish

Let me add one more honest wish. I hope my course sells well. That is an instructor’s honest wish. It is the same thing as the reason I wrote the book. So that more students will listen.

Hands

The child who read the numbers on number plates came the long way round, and round again, to here. From Daejeon to Seoul, from Seoul to Cambridge, from Cambridge to a grade-10 classroom, from the classroom back to the exam hall, and on to the stage in December. There was never a straight road. Every detour was the road. Mathematics is not a talent. It is a method. You learn that method by working through problems yourself. It can become second nature to you too.